BenaaOps (■■■ ■■■■■■■)

Saudi Civil Construction Management Portal — Technical & Functional Specification

BenaaOps is an enterprise-grade civil engineering and site operations web portal specifically engineered for general contractors, civil site engineers, and project consultants operating in the **Kingdom of Saudi Arabia (KSA)**. The system governs physical civil and structural works from deep excavation to superstructure and wet trades, ensuring full compliance with the **Saudi Building Code (SBC 301–305)**, Gulf hot-weather concrete pouring restrictions, ASTM testing protocols, and bilateral Arabic (RTL) / English (LTR) site management.

Framework:	Next.js 16 (App Router) + TypeScript	**UI/UX Aesthetic:**	Light Enterprise Engineering (Tailwind CSS)
Standard Compliance:	SBC 301-305, ASTM, ACI 305R	**Language / RTL:**	Dual EN (LTR) & AR (RTL) Toggle
Target Region:	Kingdom of Saudi Arabia (Riyadh, Jeddah, NEOM)	**Currency / Dates:**	SAR (Saudi Riyals) / Dual Hijri & Gregorian

1. Technology Stack & Light UI/UX Design System

Built according to the user's explicit directive for an **attractive, clean, and highly professional Light UI/UX** (strictly avoiding dark themes). The interface incorporates:

- **Light Foundation:** Soft warm-slate background (#f8fafc), pure white surface cards, refined hairline borders (#e2e8f0), and elevation shadows for exceptional visual hierarchy.
- **Civil Safety Palette:** Architectural Amber/Gold accents (#f59e0b, #d97706) representing Saudi construction and safety heritage.
- **Native RTL Engine:** Seamless English to Arabic (RTL) toggling dynamically modifying directionality (dir="rtl") and updating civil engineering nomenclature across all data tables and modals.
- **Dual Calendar Engine:** Simultaneous Gregorian and Umm al-Qura Hijri date display (e.g., 19 Shawwal 1447 AH / 10 Oct 2026).

2. Detailed Functional Modules Breakdown

Module 1: Executive Dashboard & Site Overview

- **4 Real-Time KPI Cards:** Active civil zones (Raft, Columns, Slabs, Masonry), Pending RFIs awaiting Consultant sign-off, Total Concrete Poured (m^3) vs project target with progress gauges, and Today's Hot Weather Safety Status.
- **Interactive Civil Milestones & Pour Timeline:** Visual stage-gate tracker tracking PCC Blinding & Tanking (100%), Substructure Raft (100%), Superstructure Columns & Slabs (64%), and AAC Blockwork (35%).
- **Hot Weather Pour Advisory & Ambient Simulator:** Interactive thermal slider (24°C to 46°C) allowing engineers to simulate ambient spikes and observe real-time SBC 304 warnings, ice batching protocols, and night pour mandates.

Module 2: Civil Phase Tracker (WBS & Progress Ledger)

Structured strictly into three Saudi building execution stages:

1. **Substructure & Deep Foundations:** Shoring secant piles, dewatering systems, PCC 1:4:8 blinding concrete, anti-termite chemical soil barrier, Bitumat 4mm tanking membrane, Sulfate Resisting Cement (SRC C40) raft, and 95% MDD Proctor backfilling.
2. **Superstructure & Frame:** Column line & plumb surveys, C45/C50 micro-silica concrete, post-tensioned (PT) strand tendon ducts, top steel clearance sign-offs, and mandatory 14-day continuous ponding curing regimes.
3. **Wet Trades & External Infrastructure:** AAC lightweight blockwork, reinforced lintels, 3-coat cementitious plastering with fiberglass mesh, polyurethane wet-area tanking with 48h ponding tests, and boundary walls.

* *Interactive Feature:* Engineers can toggle any milestone checkbox to dynamically recompute phase completion percentages.

Module 3: KSA QA/QC & Inspection Engine (RFI / WIR / MIR & Lab)

- **Work Inspection Requests (WIR / RFI):** Sequence generation (e.g. RFI-CIV-2026-1042), multi-criteria filtering by Code A (Approved), Code B (Approved with Comments), Code C (Rejected), and Pending Consultant Sign-off.
- **Official Consultant Inspection Drawer:** Modal drawer simulating the official consultant approval docket with inspector notes, signatures, and stamps from leading Gulf consultants (Dar Al-Handasah, Khatib & Alami, Zuhair Fayez).
- **Material Inspection Requests (MIR):** Tracking Mill Test Certificates (MTC) for SABIC Grade 60 and Al Rajhi rebar, ready-mix batch designs, and Bitumat waterproofing membranes.
- **ASTM / SBC Concrete Lab Cube Log:** Records Slump ($120\text{mm} \pm 25\text{mm}$), fresh mix temperature ($\le 32^\circ\text{C}$ limit), 7-day compressive breaks (70% threshold), and 28-day design strength compliance.

Module 4: Concrete Pour Scheduler & Hot Weather Logistics (SBC 304)

- **Hot Weather Thermal Protocol:** Strict enforcement of SBC 304 placement limits. Alerts when daytime casting must shift to night windows (20:00 - 04:00) with chilled water and flake ice batching.
- **Interactive Pour Booking Engine:** Booking modal with volume input (m^3), mix designs (C30, C35, C40, C50), placing equipment (Boom Pump 42m/56m, Static Line Pump), and a live **Flake Ice Replacement Slider** dynamically predicting delivery temperatures.
- **Printable Pour Permit Ticket:** Official site permit viewer complete with mix specifications, testing criteria, QC sign-offs, and print stylesheet.

Module 5: Bar Bending Schedule (BBS) & Rebar Wastage Log

- **Diameter Allocations:** Comprehensive tracking across standard diameters: $\Phi 8\text{mm}, \Phi 10\text{mm}, \Phi 12\text{mm}, \Phi 16\text{mm}, \Phi 20\text{mm}, \Phi 25\text{mm}, \Phi 32\text{mm}$.
- **Wastage Tolerance Ledger:** Tonnage reconciliation comparing Delivered vs Fabricated vs Cut-off Scrap against the Saudi industry benchmark ($\le 3.5\%$ scrap tolerance). Current portal metric: **2.68%** (Compliant).
- **On-Site Rebar Calculator:** Practical tool calculating nominal weights using standard $D^2/162.2$ formula, including tension lap splices (50Φ) and total tonnage requirements.

3. Pre-Configured Saudi Project Datasets

Project Name	Location & Type	Consultant & Contractor	Budget (SAR)
Riyadh Commercial Tower (KAFD Perimeter)	Riyadh, KSA High-Rise C50 Concrete Frame	Dar Al-Handasah Al-Bawani Contracting	SAR 245,000,000 68% Progress
Jeddah Marina Villas (Obhur Al Shamaliyah)	Jeddah, KSA Coastal Marine Dewatering & SRC	Khatib & Alami Nesma & Partners	SAR 112,500,000 76% Progress
NEOM Logistic Hub (Zone 4 Industrial)	Tabuk / NEOM Special Zone Precast & Heavy Foundations	Zuhair Fayez Partnership Saudi Binladin Group	SAR 380,000,000 40% Progress

4. Repository Links & Local Execution Guide

- **Official GitHub Repository:** https://github.com/prince-84/construction-civil.git
- **Local Dev Server:** Run npm run dev (available at http://localhost:3000).
- **Production Build:** Run npm run build (pre-compiled with Turbopack, 100% type-safe).

Prepared For: Civil Engineering & Site Operations Division	**Standard:** Saudi Building Code (SBC 301-305)

Verified By: Eng. Waleed Al-Shehri (Chief Civil Inspector)	**Date:** October 2026 / Shawwal 1447 AH